

Work Experience

Software Dev. Junior R&D Department	CETEC Structural Engineering Remote	Dec 2025 – Present
- I maintain Nexus, handling bug fixes, performance improvements, and new features, and manage the team's development: reviewing PRs to ensure adherence to project architecture and standards, requesting changes and approving merges into the integration branch; I also administer the repository and production branch, resolving conflicts, handling deployments and monitoring logs and production system health. - Led the architecture and full-stack development of Nexus, a monolithic internal management system built on a three-tier architecture (React/TypeScript, Django REST Framework, PostgreSQL) that digitizes the company's administrative, financial, and engineering processes, covering authentication and permission-level access control (including hardware-bound login for desktop plugins), a timesheet module that replaced third-party software used by contractors, a financial module with automated invoice billing and expense reporting, and a commercial workflow for converting approved proposals into engineering projects with tracking dashboards. - Integrated and orchestrated the involved technologies, deploying containerized applications with Docker on a VPS running Ubuntu Server 24.04, using Nginx as a web server and reverse proxy.		
Software Dev. Trainee R&D Department	CETEC Structural Engineering Remote	Jul 2025 – Dec 2025
- Developed plugins for Revit using its native API with C#, WPF, and Windows Forms, automating processes in structural engineering project development. - Led the initial development of plugins for Robot Structural Analysis using its API in C#, contributing to the automation and integration of engineering workflows.		
Software Dev. Junior IT Management	UniFIP (Patos University Center) Patos-PB, Brazil	Mar 2025 – Jul 2025
- Responsible for maintaining monolithic systems and APIs developed in Django, including features such as diploma generation, class scheduling, and lesson planning. Tasks included bug fixing, continuous improvement, and implementing new requirements. - Contributed to the implementation of new systems integrating legacy data, such as scheduling history, enabling the automated generation of teaching activity records.		
Software Dev. Trainee IT Management	UniFIP (Patos University Center) Patos-PB, Brazil	Sep 2024 – Feb 2025
- Designed an internal extension system between departments using Django, integrated with Docker for environment standardization and PostgreSQL as the relational database. - Developed prototypes of real-time chat systems using Django and WebSocket, and performed interface optimizations using JavaScript, Bootstrap, HTML, and CSS. - Automated academic processes through desktop applications built with Tkinter and Selenium, improving operational efficiency and reducing manual tasks for staff.		

Main Projects

Development of the Intensio Bachelor's Research

2023

- Used Python with libraries such as NumPy, SciPy, and Tkinter for statistical processing of time series data, development of graphical user interfaces, installer, and user manual, as well as supervised machine learning modeling. The project resulted in a registered computer program and a publication in the COOPEX journal.

- Used Python and libraries such as NumPy, Scikit-learn, Pandas, and TensorFlow to compare machine learning methods with traditional statistical approaches for spatial interpolation of rainfall data. The study resulted in a publication in the IEEE Access journal.

Education

- **M.Sc. Information Technology**, *Federal University of Rio Grande do Norte, Brazil.* **2024 – 2026**
- **Specialization in Structural and Foundation Engineering**, *Patos University Center, Brazil.* **2022 – 2024**
- **B.Sc. Civil Engineering**, *Patos University Center, Brazil.* **2019 – 2023**

Technologies

- **Languages:** Python, C#, JavaScript, TypeScript;
- **Libraries and Frameworks:** NumPy, Pandas, Scikit-learn, TensorFlow, Selenium, React, Django, Node.js;
- **Supporting Technologies:** Docker, Git, PostgreSQL, HTML, CSS, Jupyter, Overleaf.